

Assignment TAI

Option 3: Sewer overflow predictions

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1 Introduction

Brussels operates a combined sewer system, which collects both wastewater and rainwater. Hence water levels rise in the sewer system during and after rainfall. To prevent overloading sewage treatment plants or causing urban flooding during intense rainfall events, excess water (which is a mixture of both rain- and wastewater) is discharged directly into local watercourses without undergoing any treatment, at so-called combined sewer overflow locations. Combined sewer overflows (CSOs) have negative environmental impacts by introducing large amounts of organic matter and pollutants into rivers, depleting oxygen levels and harming aquatic life.

CSO occurrences depend on the rainfall, as well as on the initial design of the sewer system. In Brussels, there are approximately 100 CSO locations, primarily discharging into the Senne River and the Brussels Canal. However, it is a small number of these locations that contribute the most to the total overflow volume. At these key locations, CSOs can be triggered by as little as 1 mm of rainfall.

This project focuses on the Lion sewage overflow location (=DO08). This location is situated downstream of the Maalbeek sewage collector, which drains the entire Maalbeek catchment. When water levels in the collector exceed 3000 mm, overflow events occur to the Senne. You can find illustrative supports of the Maelbeek catchment location and on the working of the overflow in the project presentation powerpoint.

2 Data

The datasets you'll use are rainfall measurements from two different rain gauges (also called pluviometers) and of sewer water levels close to the Lion overflow.

Rainfall data is provided at a 5-minute resolution for the years 2022 to 2024. The data is stored in a folder named "pluviometer data", containing three Excel files, one for each year. Each file includes rainfall measurements from two stations: Pluv. Avant-Port (P01 or U24), located at the overflow point, and Pluv. Flagey (P14 or C12), situated upstream in the Maalbeek catchment. This data has been validated so that unrealistic values were removed or replaced.

Sewer level data is recorded at a 1-minute resolution and is made available for the same period (2022–2024). These records are stored in the "sewage data" folder as three CSV files, one per year. The recorded values represent the water level (in millimeters) in the sewage collector at the Lion overflow location (DO08). This is the raw data recorded by the height sensor. Be aware that this sensor is located in a hostile environment (humidity, debris in the water, high peak flows) which can sometimes make the data unreliable.

3 Assignment

This assignment consists of two consecutive tasks:

1. Predict future sewer water levels in the sewage collector next to the "Lion" overflow (i) 15 minutes ($t + 15$) and (ii) 1 hour ahead ($t + 60$).

- Apply all steps as seen during the classes and WPOs.
- Your input features should be based only on current (t) and past ($<t$) rainfall measurements and sewer levels. Consider creating additional input features to capture the temporal patterns of the baseflow.
- Apply appropriate preprocessing techniques, with a brief explanation justifying the choices you make.
- Be careful about how you split your data into the train, validation and test sets, avoid data leakage.
- Try at least three different ML algorithms, which do not belong to the same family of algorithms (e.g., using both random forest and decision tree models would not be acceptable as they are both decision tree-based). Start with simple models, those covered in class and WPOs are sufficient for this project.
- Use relevant metrics to evaluate and compare the models. You can use the ones seen in class, and if you want to go a step further you can also use Nash-Sutcliffe efficiency (NSE), a very common metric used in hydrology. Make sure to comment on how the model performance changes when predicting 15 minutes ahead vs 1 hour ahead.
- Compare your overflow detection performance to a baseline persistence model assuming steady-state.

2. After you selected the best sewer level prediction model, evaluate how well your models can anticipate overflow occurrence (water level > 3000 mm). Overflow detection should be derived from the predicted water levels, **do not treat this as a separate classification task**. Use relevant metrics to assess overflow prediction. Be aware of class imbalance.

3.1 Submission

Information regarding deliverables and deadlines can be found in the project presentation Power-Point. Check out the rubrics associated to each deliverable to know how they are graded, so you make sure you pay attention to all important elements! If needed, you may split your code into multiple files (e.g., .ipynb or .py files), however make sure the structure is logical and clear.

Pay attention to submit your deliverables using the correct title formats, we will be intransigent on this:

- TAI_code_{project number}_{first name}_{last name}
- TAI_presentation_{project number}_{first name}_{last name}
- TAI_discussion_{project number}_{first name}_{last name}